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Building a Robust Reservoir Model through Full Integration Workflow of Static and Dynamic: A Case Study from a Lower Cretaceous Carbonate Reservoir of a Giant Oil Field, Onshore Abu Dhabi, UAE

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Abstract

An updated geological and dynamic model for a giant Middle East carbonate reservoir was constructed and history matched with the objective of creating an alternative model which is capable of replicating the reservoir production mechanisms and improving predictability, allowing optimizing the field development plan and water injection strategy. Giant Middle East carbonate fields often have long production history and exhibit high reservoir heterogeneity. It is always challenging to get a robust history matched model aligned with geological concepts and dynamic behavior understanding.

The objective of this paper is to present an improved and integrated reservoir characterization, modeling and history matching procedure for a giant Lower Cretaceous carbonate reservoir in the Middle East. The applied workflow integrates all available geological data (stratigraphy, depositional facies, and diagenesis), petrophysical data (RCA and minipermeameter data, Petrophysical Group definition, cut-off definition) and the extensive database of dynamic data (long production history, well test, RST, open-hole log saturation over more than 40 years of development drilling, and MICP). The process was initiated with the reservoir review by means of a fully integrated study that allowed having better understanding of the reservoir behavior and production mechanisms. The key heterogeneities (high permeability and intra-dense layers) which control the flow behavior were identified during this process. Geological trend maps were generated to control the distribution of high permeability and intra-dense in the model. Well test data, open-hole logs from development wells and time-lapse saturation logs from observation wells were used to calibrate the trend and permeability log data. A phenomenological model was constructed to test the main factors impacting the production mechanism as identified during the reservoir review. Multiple iterations were performed between the static and dynamic models in a way that allowed a quick and efficient work that is consistent with all disciplines assumptions.

Such continuous loop between the dynamic and geological models, with focus on the geological heterogeneities driving the dynamic reservoir behavior, has led to a more robust model capable of replicate the production mechanisms, which represents a major improvement compared to previous model in term of predictability.

Introduction

Carbonate reservoirs are often extremely heterogeneous, which causes poor sweep efficiency due to complex flow paths. Very often it is found that in giant carbonate reservoirs with long production history, it is difficult to have a robust history matched model aligned with the geological and production mechanism understanding. Consequently, the optimization of the field development plan including water injection strategy can be very challenging. There are many reasons for this and some of them are the lack of understanding of reservoir heterogeneity and the insufficient integration between static and dynamic data in the model.

The objective of this paper is to present an integrated workflow of the 3D static and dynamic models, which allows replicating the dynamic behavior and production mechanisms of the reservoirs. The applied workflow integrates all available geological data (stratigraphy, depositional facies, and diagenesis), petrophysical data (RCA and miniperm-meter data, Petrophysical Group definition) and the extensive database of dynamic data (long production history, well test, RST, open-hole log saturation over more than 40 years of development drilling, and MICP). By means of this integrated reservoir review large amounts of field data including pressure, production/injection data, and time lapse saturation are also considered which give us better understanding of the dynamic behavior. This workflow was established, implemented and optimized based on the data available.

The present paper outline is as follows. Start with the overview of the studied field and general reservoir description. The next section discusses the study road map and the workflow applied to it. The integrated process from reservoir characterization into dynamic modeling is described in the following chapter. Then the history match process and the applied sensitivities to the model are discussed in a new section and finally the conclusion is presented.

Reservoir description

The carbonate reservoir under study is Lower Cretaceous in age and is part of a field situated in onshore Abu Dhabi, UAE. This field is a faulted elongated doubly plunged anticline with the longer axis trending roughly northeast-southwest. The field dimension is about 30 km long by 10 km wide, with a closure area of approximately 280 sq. km at the top of reservoir. The reservoir thickness ranges from ~200ft on the crest to ~150ft on the flank. The difference in thickness between crest and flank is largely due to the impact of compaction and diagenesis processes (cementation and stylolization) differences.

The studied reservoir was deposited in an extensive, low to moderate energy, low-angle ramp setting that stack into an overall shallowing-upward carbonate depositional sequence. It is a layered system where strata measuring a few feet in thickness can be correlated field-wide. For each well, regionally based published frameworks (Sharland et al., 2001, van Buchem et al., 2011, Strohmenger et al., 2006) and the vertical stacking pattern of the depositional facies (i.e. shallowing and deepening trends), have been used to infer a sequence stratigraphy framework. The 3rd to 5th order sequence stratigraphic framework in the reservoir formation was used to constrain the stratigraphic modeling.

The studied reservoir consists of two zones, the Lower and Upper zones, which exhibits a high contrast of permeability. The Lower zone has relatively low permeability, ranging from 1 to 20 mD, with limited heterogeneity. Whereas the Upper zone is characterized by high permeability layers ranging from 10 to 5000 mD interbedded with low permeability layers (1-100mD). The porosity in the reservoir ranges from 10 to 34%. There is no clear trend in the porosity-permeability relationship, particularly in the Upper zone; a given porosity may correspond to a permeability value ranging 2 to 3 orders of magnitude. The reservoir also contains denser intervals corresponding to thin units (1 to 3 ft thick) which, are called intra-dense layers and are explained either by early diagenetic processes and replacive dolomite rhombs associated with burrowed sediments or to late diagenetic processes linked to mechanical compaction and development of stylolite. The Hydrocarbon Initial in Place is split approximately 45% in the Upper zone and 55% in the Lower zone.

The reservoir contains light oil with the average API being 40 degree. The field has been under production for more than 40 years with the development scheme based on horizontal well infill drilling and peripheral water injector.

High contrast of permeability between Upper and Lower zones (could be up to 100 times) with some degree of communication between layers, leading to water over-ride and reverse coning (water injected into lower reservoir moves to upper reservoir and producers in lower reservoir draw water from upper reservoir) results in poor sweeping of the Lower zone. To date, the development strategy to recover the Lower zone bypassed oil is through the tighter infill drilling.

Study Roadmap

This study objective is to have an improved history matched model which is capable of replicating the reservoir production mechanism and improves predictability to assess alternative water injection strategy. The sector model was constructed within the full field model covering an area of interest for revised water injection strategy. The legacy model outside of the sector was kept to have boundary conditions representing the global flows/pressure trends around the sector. The permeability in the sector was redefined through an integrated approach.

The sector model is about 7km along the dipping direction and 5km perpendicular to it and is located in the Eastern part of the reservoir. It contains 142 wells of which 82 are horizontal and 60 are vertical. There 14 cored wells and 15 vertical well tests available in the sector model. Figure 1 illustrates the location of the sector model.

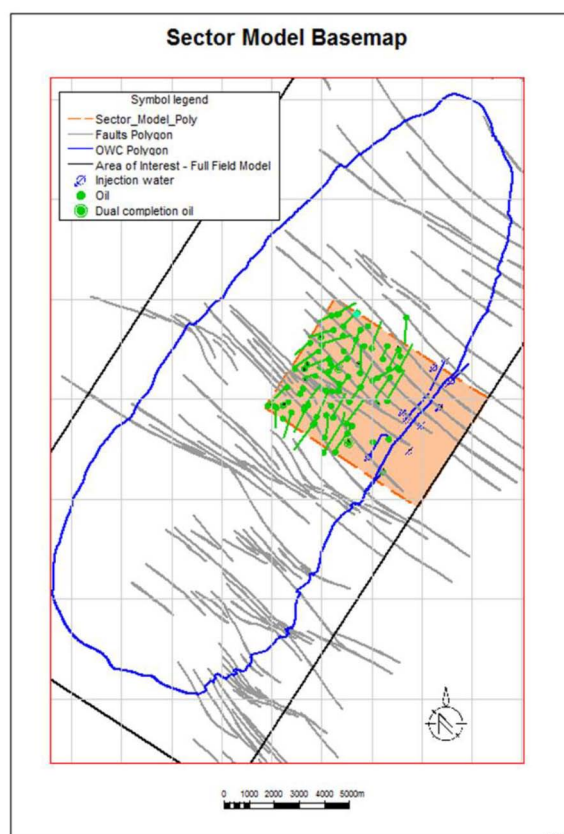


Figure 1—The sector model location with wells in respect to the full field model

The workflow was initiated by the reservoir review through a fully integrated study that allowed having a better understanding of the reservoir behavior and production mechanism.

A phenomenological model was constructed to test the main factors impacting the production mechanism as identified during the reservoir review.

The layering schemes applied in the sector model were reviewed with the objective to have common layering schemes between the static and dynamic models without overlooking the heterogeneities in the model. Calibration of core, log and dynamic data (well test) was performed to ensure the model properly captures reservoir heterogeneities such as high permeability intervals. The permeability model was redefined considering the geological understanding and dynamic synthesis. The feedback loops from the dynamic and static model were conducted and played key role in the model improvement. Multiple iterations were performed between the static and dynamic models allowing a quick and efficient workflow that is consistent with all disciplines assumptions. Figure 2 is showing the schematic workflow used in the study.

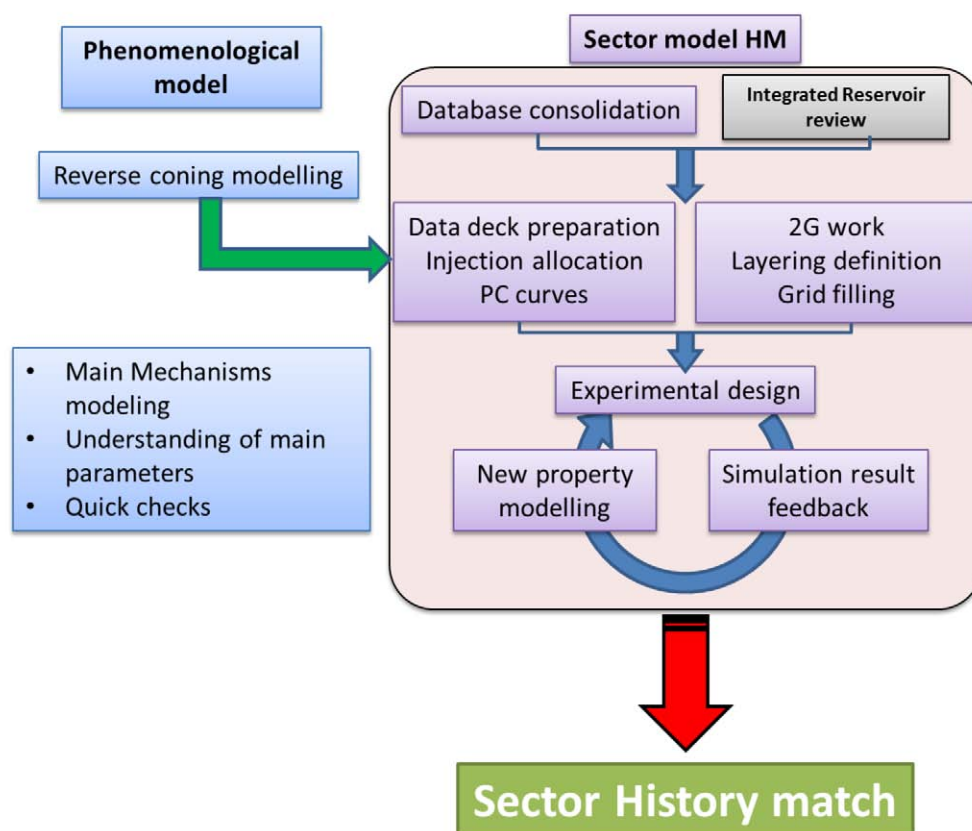


Figure 2—A schematic workflow in this study

Integrated Approach for permeability model

Integrated field reservoir review

The process of the study was initiated with the field reservoir review by means of a fully integrated study aiming to better characterize of the reservoir behavior and production mechanism which illustrated in the Figure 3. This integrated reservoir review involved using large amounts of field data including pressure, production/injection data, and time-lapse saturation. The main geological heterogeneities controlling the flow behavior were identified during this process. This included the identification of all areal and vertical variations in reservoir properties that can affect the fluid flow (accelerates and relaxes) as well as, the proper characterization of the heterogeneities which impact the production mechanism.

The time-lapse saturation logs were reviewed and assessed to understand the sweep efficiency and water evolution in this highly heterogeneous reservoir. They contain information on changing saturation,

sweep efficiency, permeability contrast between Upper and Lower zones, and the remaining oil saturation distribution. Pressure Transient Analysis was conducted to review areal characterization. The heterogeneity between crest and flank is seen on PTA. However, there is no clear trend between the North and South part of the field in the oil pool; the Kh values are quite similar between those regions. The Kh data from PTA in the highly faulted area does not show any different trend than the rest of the wells. RFT analysis and interference test were used to review the vertical connectivity performance showing that the intra-dense layers within the reservoir are more likely to act as baffles rather than as barriers. The formation pressure testing (MDT and RST) data shows that pressure communication exist between the Upper and Lower zones. The laboratory test study on stylolite zones demonstrated that the displacement across a stylolitic zone is possible at the plug and whole core scales.

Based on this review, the production mechanism can be defined as follow and illustrated in Figure 3:

1. Peripheral Lower zone water injection override to Upper zone; driven by the permeability contrast between Upper and Lower zones
2. Water hold-up in Upper zone due to permeability contrast and presence of capillary pressure barrier
3. Rapid water front movement in Upper zone caused by the presence of high permeability streak in Upper zone and K_v/K_h
4. Water slumping (reverse coning), triggered by the production from Lower zone wells and kinetic driven by intra-dense layers and K_v/K_h .

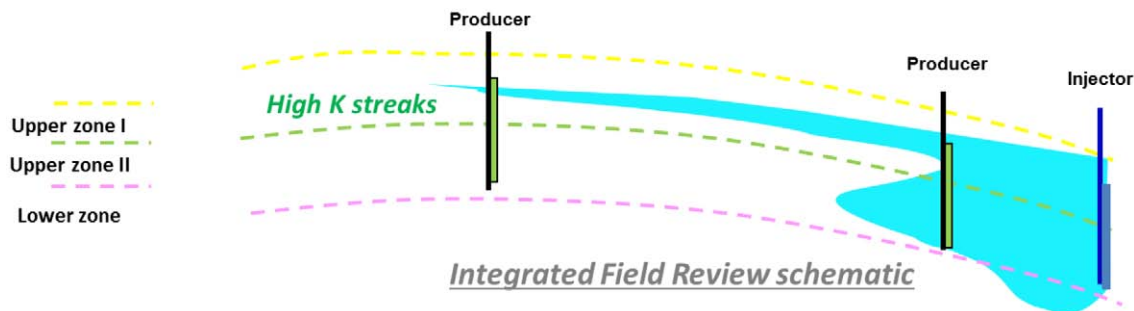


Figure 3—Schematic of reservoir behavior and production mechanism define during Integrated Field Review.

Phenomenological model

The objectives of the phenomenological model is to check the main factors which impact the production mechanism identified in the integrated field reservoir review. A slice model was taken from the full field Thamama model and tried out both in fine and coarse layering using simplified development scenarios containing few wells.

During this analysis, it was observed that the water over ride can only be reproduced if there is a high permeability contrast between Upper and Lower zones. The difference in Pc imbibition curve between the most dominant rock type in Lower zone and Upper zone could explain the water hold-up phenomenon in Upper zone. Therefore the filling of the rock-type in the model will define the way water will sweep into the Lower zone. The phenomenological model also indicated that the capillary pressure has a big impact on determining the amount/kinetics of trapped oil; therefore it is essential to consider the sensitivities on the capillary Pressure/Pc (Figure 4).

The phenomenological model helped to focus rapidly on the keys elements affecting the history match process to achieve the objective to replicate the dynamic behavior and production mechanism of reservoirs.

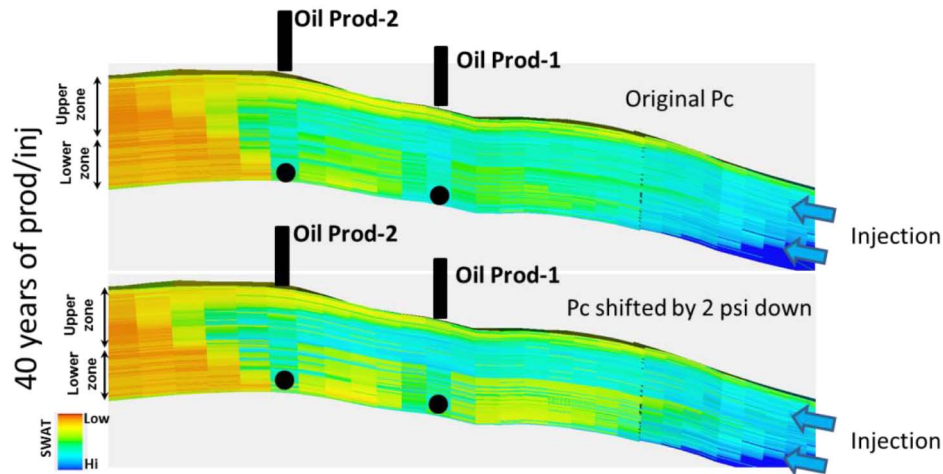


Figure 4—Snapshot after 40 years in the Phenomenological model, showing the impact of P_c on the determining the amount/kinetics of trapped oil in Lower zone.

Permeability model update

In the legacy model, lack of energy leads to adopt the following concept in the history match. An attempt to use permeability multipliers was done along the fault corridors with the assumption of having fractures along the faulted areas. This approach with the additional deterministic process of modifying the intra-dense layers (to control the permeability barrier at interface between the Upper and Lower zones) and cutting the water injection rate allowed to have an acceptable the history match. However the model could not replicate the dynamic behavior and production mechanism observed in the reservoir (reverse coning). Moreover, the fracture study conducted recently has shown that the fracture is not the main driver for flow performance. The latest integrated reservoir review also did not support the assumption applied in the legacy model of having a permeability barrier at the interface between the Upper and Lower zones to control the water hold-up phenomenon.

In the new model, the workflow was strongly driven by the requirement for the dynamic simulation. The implementation was done through an integrated and iterative process that included revised geological concepts, dynamic data, dynamic synthesis and feedback from dynamic simulation. In the following section will present a brief description of the different steps to update the permeability model. Figure 5 illustrates the integrated workflow for the permeability model update, which will be discussed further in the following section.

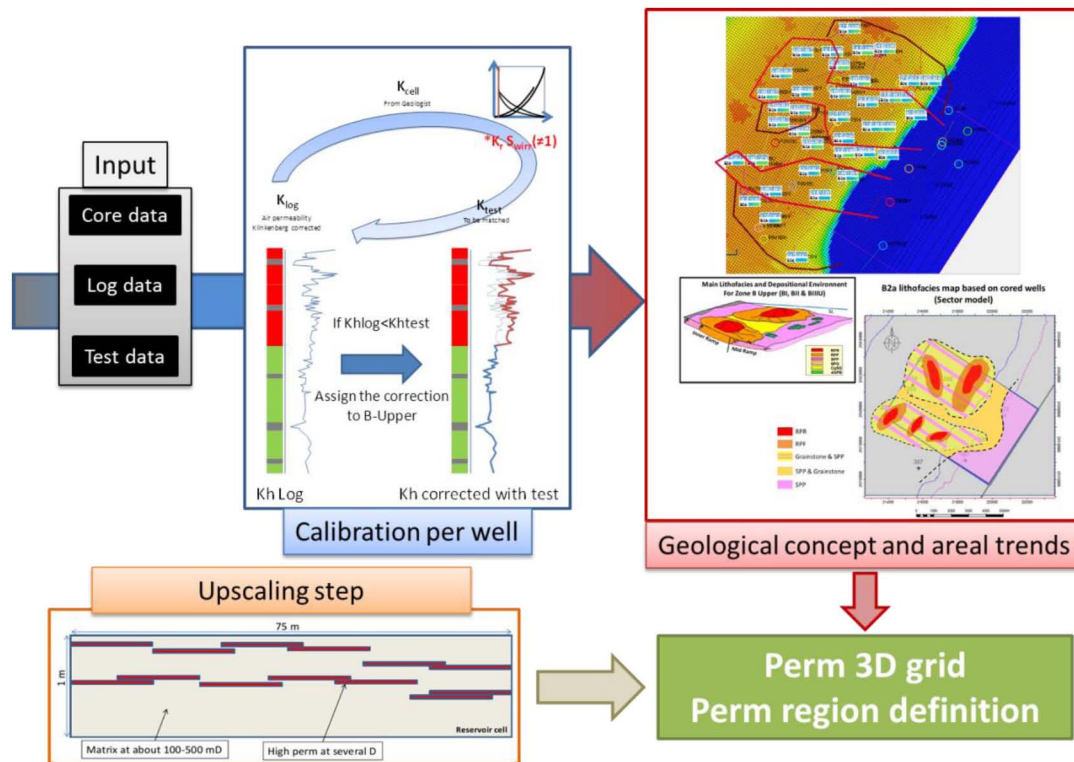


Figure 5—A schematic integrated workflow for permeability model update

Well Test Calibration. The first approach in the permeability model update is to calibrate the static data and dynamic data. Calibration of core, log permeability and well test data was carried out to ensure the absolute permeability values are represented in the small scale data (well logs data) which later will be used as input in the modeling. Masalmeh, et al. (2012), clearly pointed out that the permeability measurement from the core data which is used to characterize the reservoirs only represents a fraction of the possible scales and corresponds to a biased dataset. Therefore, a careful integration of permeability data from the plug to the well scale, going back to core description is crucial to obtain a realistic model. Figure 6 illustrates example of closer core investigation indicating how the core plugs data underestimates the permeability values due to a biased plug selection. The high permeability intervals that are often are only a few centimeters thick were missed in the standard sampling procedure, therefore the core data could significantly under represent the heterogeneity present in the reservoir. Minipermeameter measurements can also be used to identify the presence of the higher permeability intervals. Masalmeh, et al. (2012) demonstrated in their study that the minipermeameter measurement taken along the core with a sampling distance of 1 cm suggested the presence of higher permeability at a much higher frequency than that encountered by the RCA data.

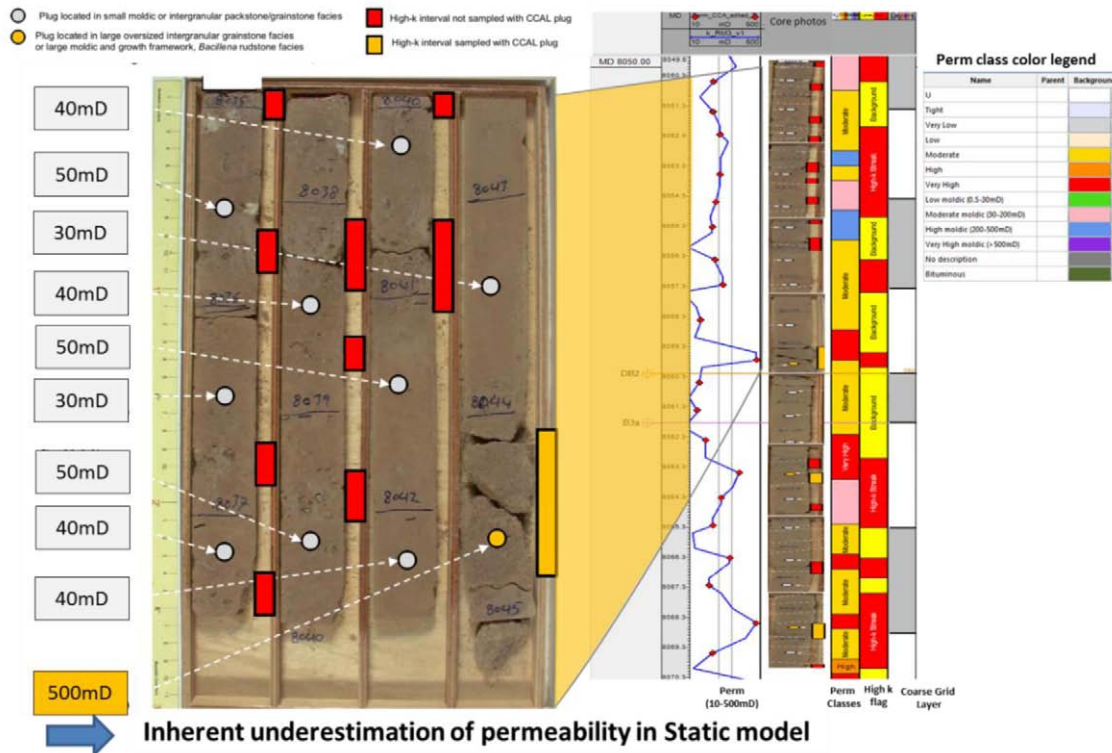


Figure 6—Example of biased core plugs data selection leading to underestimating the permeability values

Trend maps. The geological trend maps were generated to control the distribution of the main heterogeneity in the model. As a result of the feedback from the integrated field reservoir review, the permeability model update focused on two main heterogeneities: the high permeability streaks and the intra-dense layers. Based on the core description, two main types of high permeability streaks have been identified; these are rudist facies and stormy type deposits of grainstone facies. High permeability streaks are characterized by well-developed intergranular porosity and local intragranular pores consisting primarily of a preserved syn-depositional to shallow burial pore network that is a result of very early to early diagenetic processes. Core and thin section descriptions, as well as the geochemical data, consistently indicate that no obvious dissolution feature is involved in the origin of the high permeability streaks. The static model distribution "laws" should be related to depositional environment maps. However, these thin layers (sometimes < 1 ft to a maximum 3-4 ft) show significant heterogeneity vertically and laterally based on the core data. Mapping the high permeability streaks is then a very challenging task, as the high permeability streaks could be present across stratigraphic layers, and it cannot be based on well correlation (Pamungkas, et.al, 2016).

A high permeability streak flag was generated in order to help identify the high permeability intervals at wells and enable to generate the proportion maps for each zone. Core permeability statistics per subzone were used to define the cut-off for the high permeability flag. Zones displaying a steep slope on the quantile distribution indicate the presence of heterogeneity and high permeability streaks. 2D proportion maps were generated based on the high permeability streak flags for each zone. Multiple scenarios of high permeability boundary were defined based on different cut-off scenarios to capture the uncertainty on the lateral extension of the high permeability streak (Figure 7). The geological concept and depositional model was used to adjust the proportion maps. Lastly, the calibration with the well test data and time-lapse saturation logs was conducted in order to align with the dynamic synthesis.

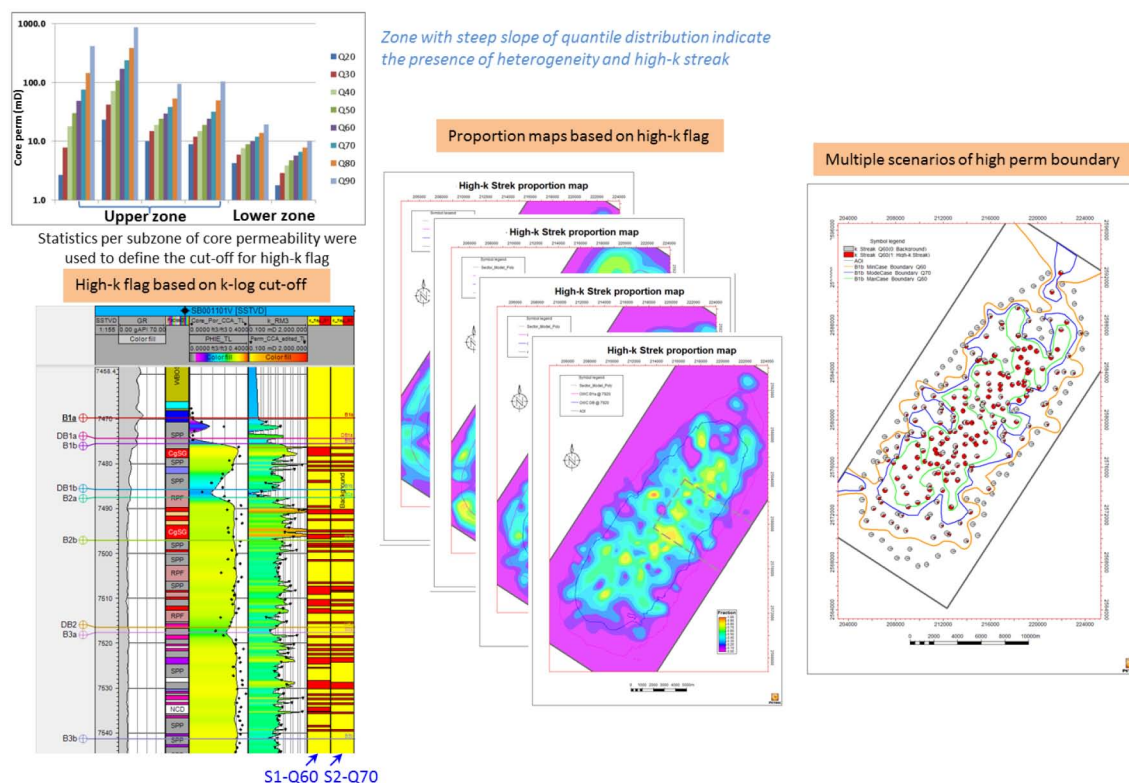


Figure 7—Generation of proportion map based on high-k flag

Layering refinement. In order to allow fast and efficient iteration loops between the static and dynamic models, a common layering scheme was applied. The new layering rationale was applied by considering the key geological heterogeneities and selecting the optimum numbers of layers that allowed the dynamic model running and preserving explicit modeling of the highest permeability. The high permeability streak flags were used to check whether the new layering schemes are capturing the heterogeneity by ensuring that no very high permeability information was lost during the upscaling process. For the intra-dense layers, as they were modeled as specific layers/sub-zones no further refinement was required.

Permeability model. In this study, the modeling strategy is to apply different approaches to model the main geological heterogeneities: High permeability intervals, intra-denses, and the background. As there is a weak relationship between poro-perm in the high permeability intervals, modeling permeability in these intervals using conventional cloud transforms or stochastic co-simulation with porosity will be misleading. Therefore, different geological heterogeneities (high permeability streaks/intra-denses) require specifically adapted modelling strategies. Trend maps for high permeability streaks and intra-dense layers were used to constrain the permeability model in those intervals. Later in the modeling, the 2D trend maps were used as horizontal trends in the Truncated Gaussian Simulation and the vertical proportion curve as vertical trend to generate the 3D proportion cubes of high permeability and intra-dense. This approach allowed body geometry and magnitude of the high permeability and intra-denses to be a key history-match parameter. In the background intervals Sequential Gaussian Simulation was used to model the permeability by co-simulating it with the porosity. The base case variogram ranges used in the full field model (FFM) were applied in the sector model.

History Match

The dynamic model was simulated in full field mode with the legacy full field model used for outside the sector model and the refined new model in the sector model. The legacy model outside of the sector model

was kept to have boundary conditions representing the global flows/pressure trend around the sector (Figure 1).

The principle applied in the simulation is to have a multi-scenario approach to be able to understand and assess the different levers for history match. Two main steps have been followed, based on production mechanism understanding:

Step 1: Upper Zone

- Match the events that drive the Upper zone quick flooding:
 - Early production / injection (1980's)
 - Water ingress by looking at Saturation logs and Well water cuts: early Water breakthrough and water front in preferred pathway, in specific sub layers
 - Pressure response: connectivity peripheral injectors to producers
- Main drivers for the history match:
 - Improved permeability model in Upper zone using well test Kh
 - Permeability with multipliers in dedicated layers using geological contours, controlled through open hole saturation data integration (efficient way to reach a geologically driven history match)
 - Understanding of upscaling of permeability values (K_v/K_h)
 - Capillary pressure: gap between upper and lower values

Figure 8 illustrates the results of an experimental design for several Upper zone water cut.

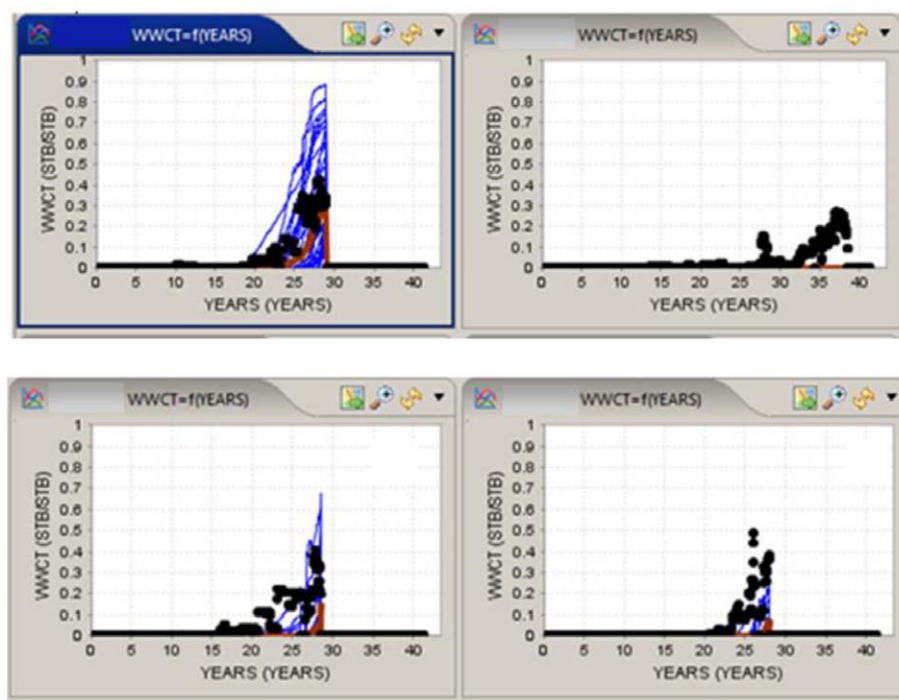


Figure 8—Upper zone - Example of Water Cut Match of an experimental design plan

Step 2: Lower zone

- Match the events that drive the Lower flooding:
 - Injection and production rates in the first years of lower production (vertical wells)

- Poor sweeping at first
- Pressure communication Upper/Lower (no barrier)
- Water ingress by looking at:
 - Water breakthrough time: due to presence of high water saturation in upper, water slumping to Lower zone triggered by Lower zone well drawdown
 - kinetic of the slumping (water cut evolution) controlled by K_v/K_h and Intradense
 - By-passed oil and cone shape through key saturation logs
- Main drivers for the history match
 - Capillary pressure barrier between lower and Upper)
 - Permeability for initial production data on vertical strings
 - K_v/K_h
 - Intradenses for reverse coning (mostly in Upper zone), using the latest understanding and trend maps
 - Relative permeability by using different but plausible Corey exponent values in High perm intervals/B-Lower

Figure 9 illustrates the results of an experimental design for several Lower zone water cut.

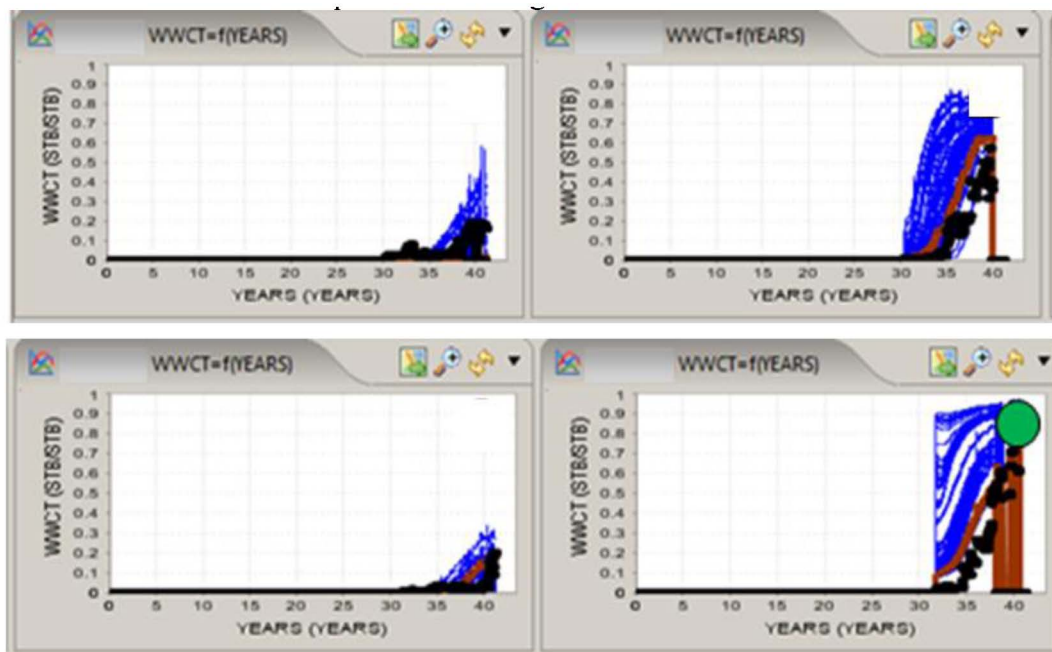


Figure 9—Lower zone – Examples of Water Cut match of an experimental design plan

For each step, a matrix of parameters (Experimental Design) following Geological contour have been generated (e.g. permeability multiplier in a given region, K_v/K_h at a given interface between Upper and Lower zone). The analysis using Objective function has been used to select best realizations, while mapping the quality of the match (water cut, saturation or pressure). Based on the analysis, iterations to the static model are performed to adjust the contour within geological understanding and re-generate regions and permeability map. In this study, nine loops back/iterations have been conducted to achieve an acceptable alignment between static and dynamic.

The feedback loop from the dynamic model into the static model plays a key role in the model improvement. Sharing the same layering schemes between static and dynamic models allowed having fast and efficient loop iteration without having to deal with the upscaling impact. This was especially important to be able to focus a small volume of water travelling very quickly through thin layers as it is observed in reality.

Iterative simulations were performed using different scenarios and various sensitivities. Several iterations with some changes on the parameters / filtering / adjustment of values led to the final batch of cases.

History match case selection

With a large number of data to match and multiple realizations from different scenarios to deal with, an objective function was generated to assess the quality of the history match every time. It quantifies the mismatch between selected relevant historical points and the simulated values and watercut/pressure/saturation data. The history match workflow also follows the understanding of the production mechanism:

- Step-1 : Breakthrough timing and water cut evolution in Upper zone
- Step-2 : Slumping and time to water breakthrough in Lower zone

During this process, a few points were selected allowing to look at the breakthrough and end value as well as the trend by tracking the watercut parameter. Depending on how close the simulated values are from the trend, a grade was given to the model. The best ones being selected are the ones with the lower grades. This approach proved to be a very efficient way of selecting models with a good match. An example of well point selection for history match is showing in the next figure (Figure 10).

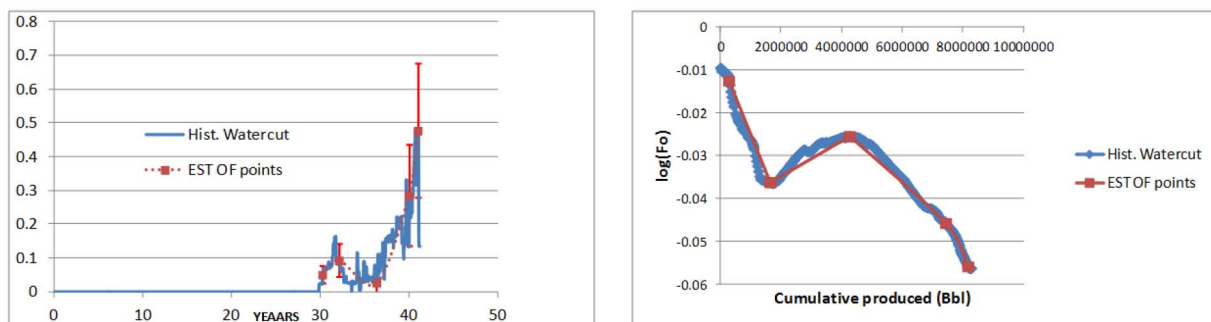


Figure 10—Example of well point selection for history match

An overall very satisfying match was obtained during this study, while honoring the production mechanism.

Figure 11 show the example of the history match in which the reservoir production mechanism is also replicated in the model.

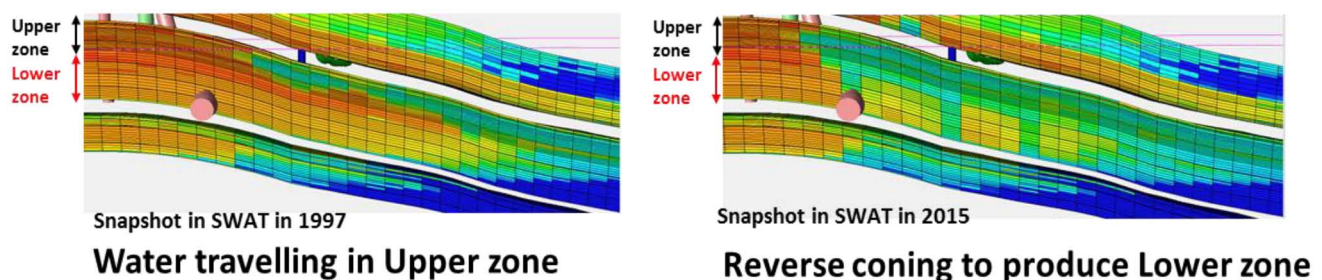


Figure 11—History matches illustration

The following figures present some examples of the quality of the history match:

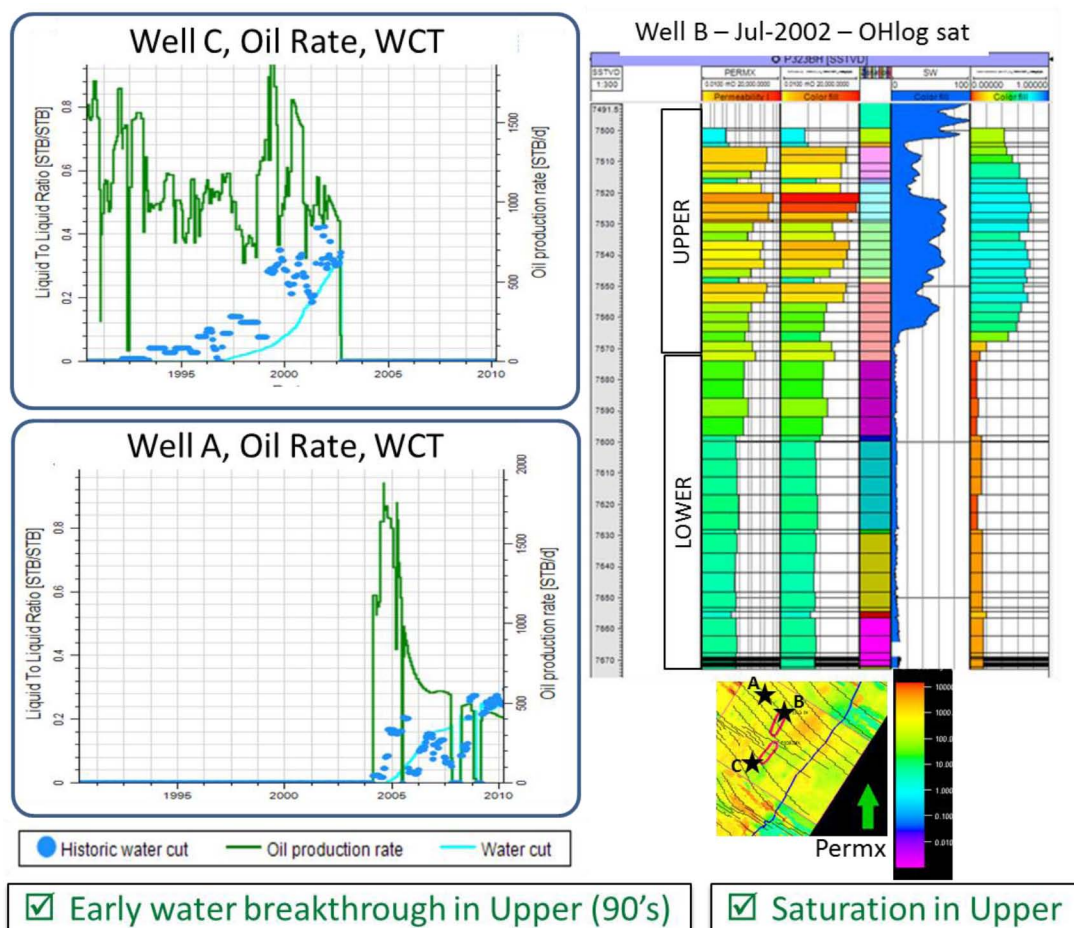


Figure 12—Example of the quality of Upper zone history match

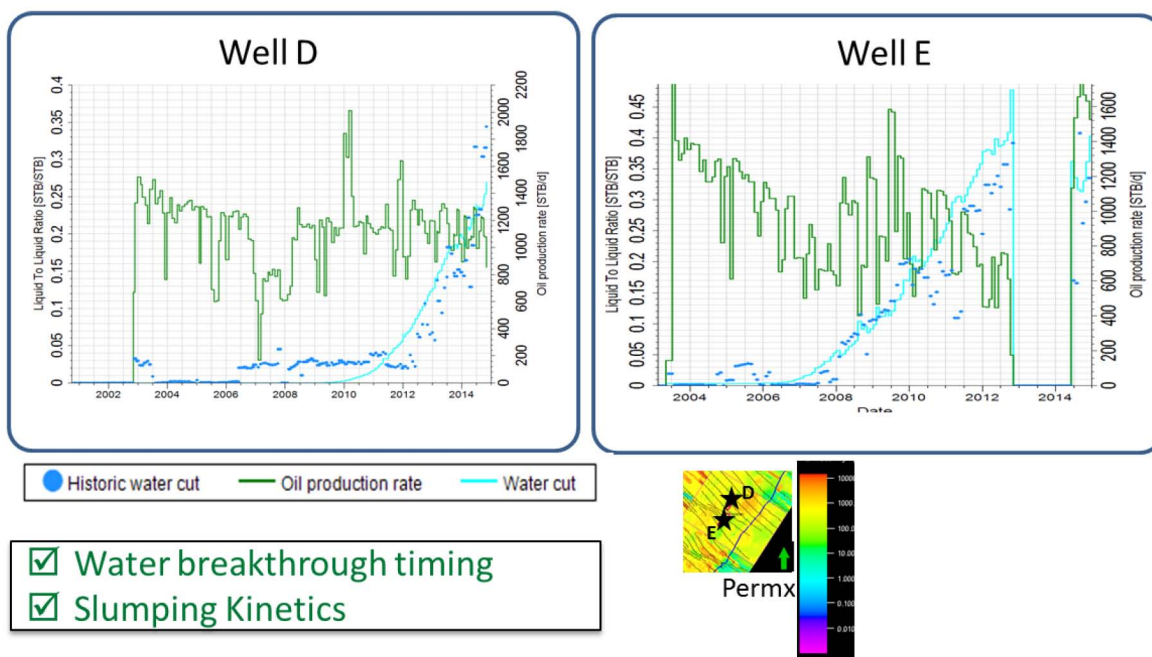


Figure 13—Example of the quality of Lower zone history match

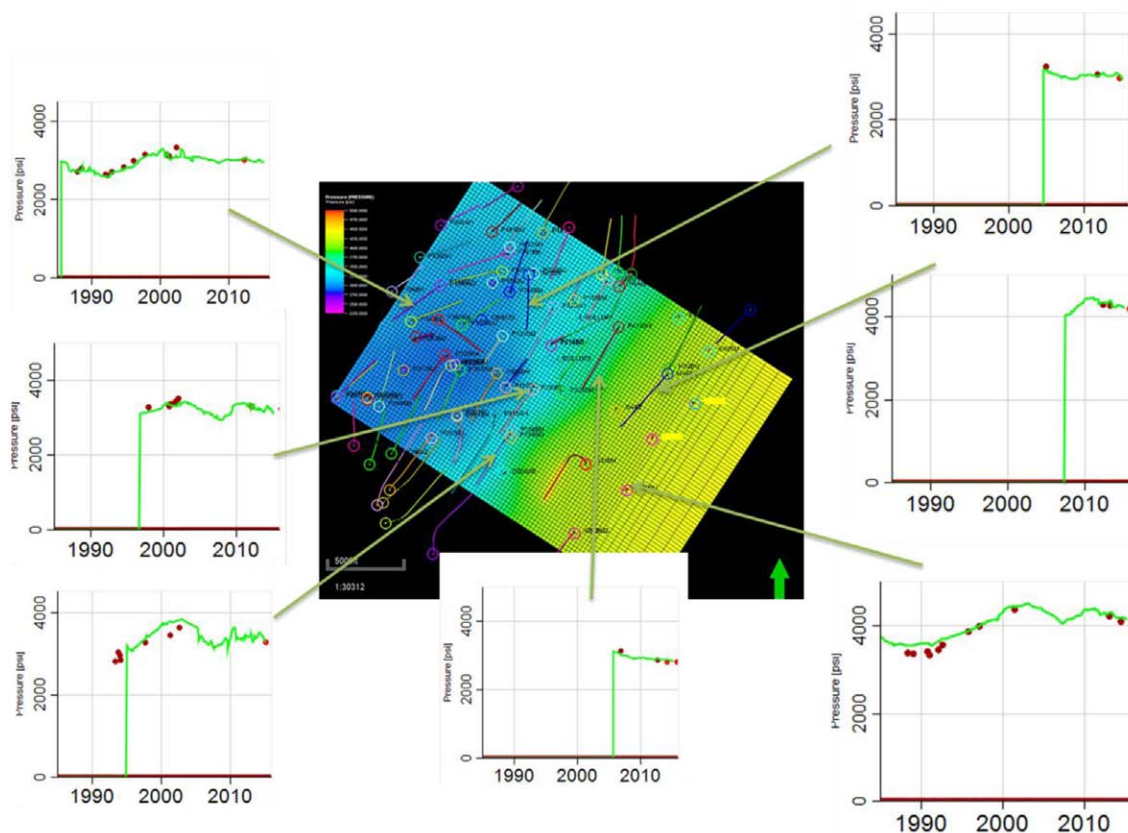


Figure 14—Example of the quality Pressure history match

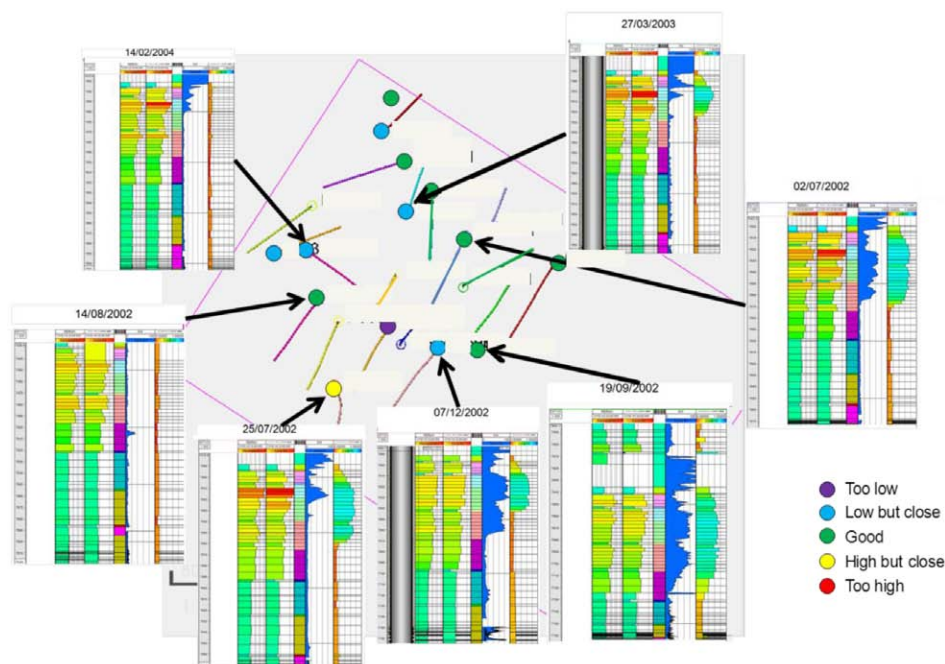


Figure 15—Example of the quality of the saturation match in the sector model

Conclusion

The integrated reservoir review helped to understand the reservoir behaviors and the main heterogeneities impacting the fluid flow. The calibration of permeability logs and well tests was required in the beginning of the process to ensure the alignment between small scale and large scale data.

The phenomenological model helped to focus rapidly on the keys elements affecting the history match process to achieve the objective to replicate the dynamic behavior and production mechanism of reservoirs.

Continuous loop back from dynamic to the static model, with focus on geological heterogeneities driving the dynamic reservoir behavior, played a key role in the model improvement. Multi-realization with experimental design allowed to screen and selects the best match model.

This applied workflow resulted in improved and more robust history match models which are capable of replicate the production mechanism. The fully integrated approach allowed a quick and efficient work with consistent hypotheses in all disciplines.

Based on the success of this approach, a similar workflow will be applied to the entire reservoir. The updated sector model is being used to assess alternative development scheme such as new water injection strategy and EOR mechanism.

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